

Your grade: 100%

Your latest: 100% • Your highest: 100% • To pass you need at least 70%. We keep your highest score.

1. Which of the following statements is NOT true about the set $A \cup B$, where A and B are both countable sets?

1 / 1 point

☐

$|A \cup B| \leq |A| + |B|$

☐

$A \cup B$ is countable

☒

$A \cup B$ is uncountable if A is countable infinite

☐

$A \cup B$ is finite if A and B are both finite

👍 Nice work

Because the union of two countable sets is always countable, regardless of whether they are finite or infinite. Thus, $A \cup B$ cannot be uncountable if both A and B are countable.

2. What is the cardinality of the set $A = \{1, 2, 3\}$?

1 / 1 point

☐☐☐☒

👍 Nice work

This set has 3 elements. So its cardinality is 3.

3. Which of the following sets has the same cardinality as the set of natural numbers $\mathbb{N}$?

1 / 1 point

☐☐☐☒

👍 Nice work

All other options are all uncountable. However, we must have a countable infinite set.

4. Which of the following is NOT true about the set relation $A \setminus B$ (i.e. means all elements in A that are not in B)?

1 / 1 point

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$A \setminus B = A - (A \cap B)$

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$A \setminus B \subseteq A$

☒

$A \setminus B = A$ if $(A \cap B) \neq \emptyset$

☐

$A \setminus B = \emptyset$ if $A \subseteq B$

👍 Nice work

$A \setminus B = A$ only if $A \cap B = \emptyset$, not if $A \cap B \neq \emptyset$.

5. Given sets A and B with $|A| = |B| = |\mathbb{N}|$, which of the following is true about $A \times B$?

1 / 1 point

☒

$A \times B$ is countable

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There can not be any bijection between $A \times B$ and $\mathbb{N}$

☐

$|A \times B| = |\mathbb{R}|$

☐

$A \times B$ is uncountable

👍 Nice work

Because the Cartesian product of two countable sets is countable. Since $|A| = |B| = |\mathbb{N}|$, and $\mathbb{N} \times \mathbb{N}$ is countable, $A \times B$ is also countable.

6. Which of the following sets is NOT countable infinite?

1 / 1 point

☐☐☒☐

👍 Nice work

This set is uncountable.

7. Assume that set A has infinitely many elements and set B has finitely many elements. Which of the following sets is guaranteed to have infinitely many elements?

1 / 1 point

☐

$A \cap B$

☐

$B - A$

☐

$P(B)$

☒

$A - B$

👍 Nice work

This set is guaranteed to have infinitely many elements.

8. Which of the following is equivalent to the set relation $A \cap (B \cup C)$?

1 / 1 point

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$(A \cup B) \cap C$

☐

$A \cap (B \cap C)$

☒

$(A \cap B) \cup (A \cap C)$

☐

$(A \cap B) \cup C$

👍 Nice work

By the distributive law of set theory, $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$.

9. Which of the following is NOT true about the set relation $A \cap B$?

1 / 1 point

☐

$A \cap B \subseteq A$

☐

$A \cap B = A$ if $A \subseteq B$

☒

$A \cap B = A$ if $A \cap B \neq \emptyset$

☐

$A \cap B = B \cap A$

👍 Nice work

This is not necessarily correct because $A \cap B = A$ only if B contains A , not merely if their intersection is non-empty. The other statements are standard properties of intersection.

10. What is the power set of $\{a, b\}$?

1 / 1 point

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$\{a, b, \emptyset\}$

☐

$\{a, b, a, b\}$

☒

$\{\emptyset, \{a\}, \{b\}, \{a, b\}\}$

☐

$\{a, b\}$

👍 Nice work